UNIVERSITY of NICOSIA

Session

DAOs, Decentralized Social Media, and Digital Identity

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Session Objectives

- Explore Self-Sovereign Identity (SSI) and how it enables individuals to control their digital identities without central authorities.
- Understand the role of blockchain in Decentralized Identifiers (DIDs) and Verifiable Credentials (VCs) for secure and private identity verification.
- Understand the Ethereum Name Service (ENS) and its role in simplifying blockchain addresses and enabling censorship-resistant naming.
- Explore ENS use cases, including cross-chain functionality, decentralized websites, and SSI integration.
- Analyze the benefits of SSI for privacy, data control, security, and fraud prevention.
- Understand Proof of Humanity (PoH) and Universal Basic Income (UBI) in decentralized identity systems.
- Examine how DAOs support decentralized governance and identity validation.
- Explore DAO case studies and their role in identity management.

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Agenda

1. Self-Sovereign Identity (SSI)
2. Ethereum Name Service (ENS)
3. Proof of Humanity
4. Introduction to DAOs
5. DAO Examples and Characteristics
6. Decentralized Social Media
7. Conclusions
8. Further Reading

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Session: DAOs, Decentralized Social Media, and Digital Identity

1. Self-Sovereign Identity (SSI)

What is Self-Sovereign Identity (SSI)?

- SSI provides individuals full control over their digital identity without relying on third-party authorities.
- SSI is often used interchangeably with decentralized identity, enabling users to self-manage their digital identities through decentralized technologies.
- **Digital Identity:** Any online data that can be linked to an individual or organization (e.g., passwords, usernames, bank accounts, social media profiles, etc.).

- **Evolution of SSI:** The journey of SSI began with the **"Web of Trust"** concept introduced by PGP in the 1990s, which decentralized trust through peer validation. In 1996, Carl Ellison's paper proposed verifying identity without traditional certification authorities. Finally, the rise of blockchain technology in the 2000s provided the decentralized infrastructure that enabled SSI to become a practical and secure solution for digital identity.

Source: https://www.dock.io/post/self-sovereign-identity

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Centralized vs. decentralized identity

Aspect	Centralized Identity	Decentralized Identity
Control	Controlled by a central authority (e.g., government, corporation).	Individuals have full control over their own identity.
Data Storage	Data is stored on centralized servers, owned by organizations.	Data is stored in a decentralized manner.
Security	Risk of breaches, hacking, and misuse of personal data.	Enhanced security through cryptography and decentralized storage.
Privacy	Users' data can be accessed and shared by centralized entities.	Good overall, dependent on root chain for dispute resolution
Ownership	The entity holding the identity has ownership of the data.	Individuals own and manage their identity and data.
Access	Users access systems through accounts managed by central authorities.	Users access systems directly, with permission granted at their discretion.
Trust	Trust is placed in the central authority.	Trust is distributed and cryptographically verified.
Interoperability	Limited, often siloed across different platforms.	Highly interoperable across platforms using standardized protocols.
Risk of Control Loss	Low risk	High risk

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SSI involves 3 key participants

1. **Holder:**
   - The Holder is the individual or entity that owns and controls their digital identity.
   - They manage their identity with Decentralized Identifiers (DIDs) and Verifiable Credentials (VCs), ensuring secure, private identification.
   - The Holder controls what data to share, with whom, and can revoke access at any time.

2. **Issuer:**
   - The Issuer is a trusted entity that creates and issues credentials validating the Holder's identity.
   - Examples include governments (ID cards, passports), universities (degrees), and businesses (employment verification).
   - The Issuer digitally signs credentials, ensuring authenticity for third-party verification.

3. **Verifier:**
   - The Verifier checks the authenticity of the Holder's credentials.
   - This could be a bank, employer, or government agency requiring proof of identity for services.
   - The Verifier does not store identity data but validates the credentials through cryptographic proof.

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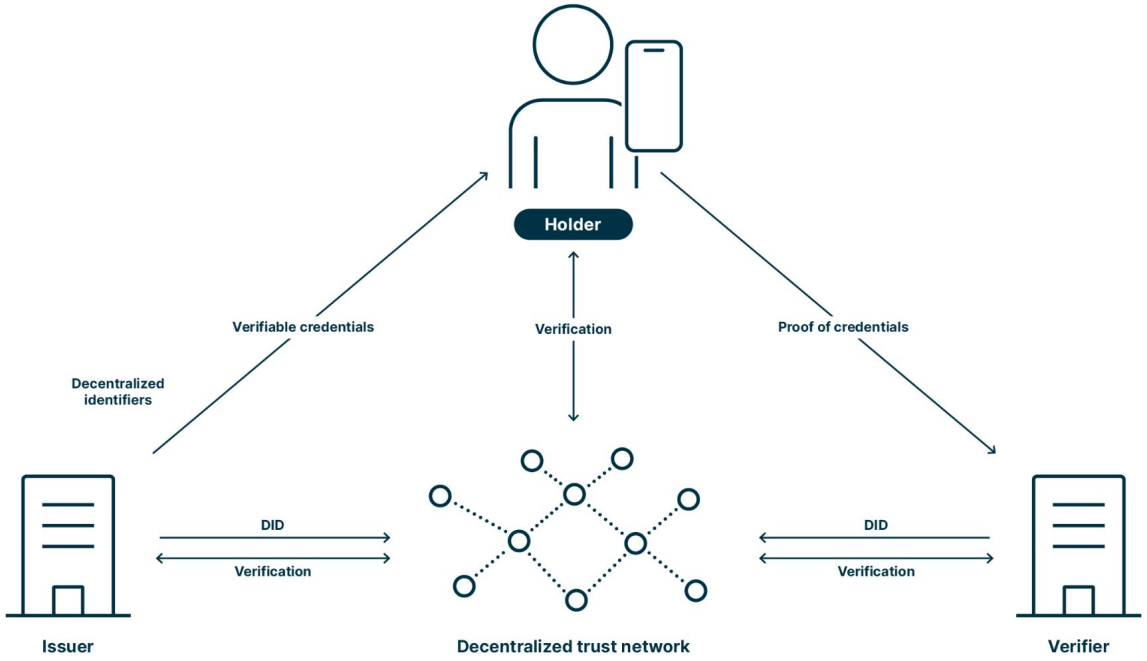
SSI data flow between its participants

The Trust Triangle Concept:

- The **Trust Triangle** consists of the three key participants (Holder, Issuer, and Verifier) working together to enable secure identity management.
- This decentralized framework empowers users to **retain control** over their data while allowing Issuers and Verifiers to validate credentials in a secure and trustworthy manner.
- By leveraging cryptography, the system ensures data integrity, privacy, and security, significantly reducing the risks of breaches and fraud compared to traditional centralized identity systems.



Source: Adnovum

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Key pillars of SSI

1. **Blockchain:** Provides a decentralized, tamper-proof ledger ensuring data integrity and preventing unauthorized changes to identity data.
2. **Decentralized Identifiers (DIDs):** Unique, blockchain-based identifiers that allow self-managed, verifiable, and unlinkable digital identities.
3. **Verifiable Credentials (VCs):** Cryptographically secure, digitally signed credentials issued by trusted authorities, which can be verified on the blockchain to prevent fraud.

These three pillars **work together** to create a secure, **decentralized identity system**. Blockchain ensures data integrity, DIDs give users control over their identities, and VCs allow secure, verifiable credential sharing without fraud risk.

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Benefits of SSI

- **For Individuals:**
  - **Ownership and Control:** Individuals can fully control their personal data without relying on centralized authorities.
  - **Privacy and Security:** Individuals can decide what data to share and with whom, minimizing risks of breaches and misuse.
  - **Portability:** Individuals can use their identities and credentials across various platforms and services.

- **For Organizations:**
  - **Cost Efficiency:** Organizations save time and resources by issuing and verifying credentials efficiently.
  - **Enhanced Security:** Cryptographic methods ensure data integrity and reduce identity fraud.
  - **Reduced Liability:** Less centrally stored user data means fewer risks of breaches, lawsuits, or fines.

- **For Developers:**
  - **Improved User Experience:** Developers create apps with password-less authentication for seamless use.
  - **Direct Data Access:** Developers can request data directly from users, bypassing intermediaries.
  - **Decentralized Solutions:** Build dApps leveraging SSI for secure identity verification.

Source: https://www.dock.io/post/self-sovereign-identity

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The 4 phases of digital identity evolution

1. **Centralized Identity:**
   - Managed by a single authority (e.g., government, company) where users have limited control.
   - **Common systems:** Email addresses, government-issued IDs.
   - **Security risks:** Centralized databases are vulnerable to hacking, data breaches, and loss of user control.

2. **Federated Identity:**
   - Identity is managed by multiple authorities, but there is still reliance on a central identity provider (e.g., Google, Facebook login).
   - This provides some convenience but still has privacy concerns, as data is stored in centralized databases.

3. **User-Centric Identity:**
   - Users have more control over their data but still interact with centralized systems and authorities to manage and authenticate their identity.

4. **Self-Sovereign Identity (SSI):**
   - A fully decentralized approach where individuals manage their identity on the blockchain, giving them complete control over their data and how it's shared.
   - SSI provides the highest level of security, privacy, and autonomy.

Source: https://www.dock.io/post/self-sovereign-identity

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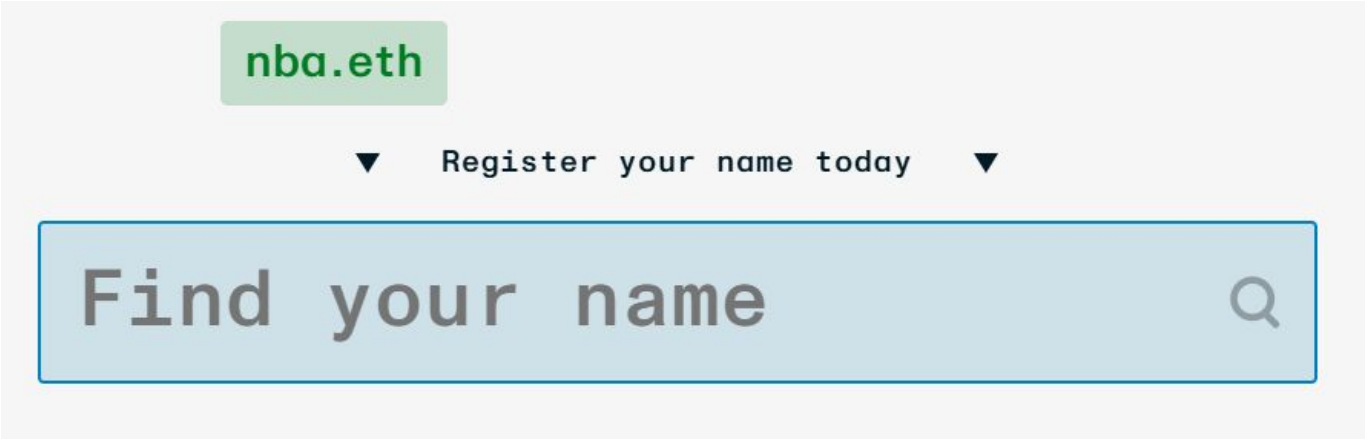
Session: DAOs, Decentralized Social Media, and Digital Identity

2. ENS (Ethereum Name Service)

Ethereum Name Service (ENS) overview

The Ethereum Name Service (ENS) is a decentralized naming system built on the Ethereum blockchain.

- ENS allows the translation of long, complex blockchain addresses into human-readable names, making interactions with blockchain networks more user-friendly.
- It is designed to support not just Ethereum addresses but a broader ecosystem, enabling cross-chain naming for various blockchain services.



Source: https://ens.domains/

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Evolution and history of ENS

- The idea of ENS was introduced in 2016 by Nick Johnson as a proposal within the Ethereum Foundation.
- The first ENS names were minted in 2017, starting with rilxxlir.eth, and the .eth Top-Level Domain (TLD) was designated for ENS addresses.
- ENS became an independent project in 2018 with the support of the Ethereum Foundation, which provided a grant to continue its development and growth.

nick.eth on X

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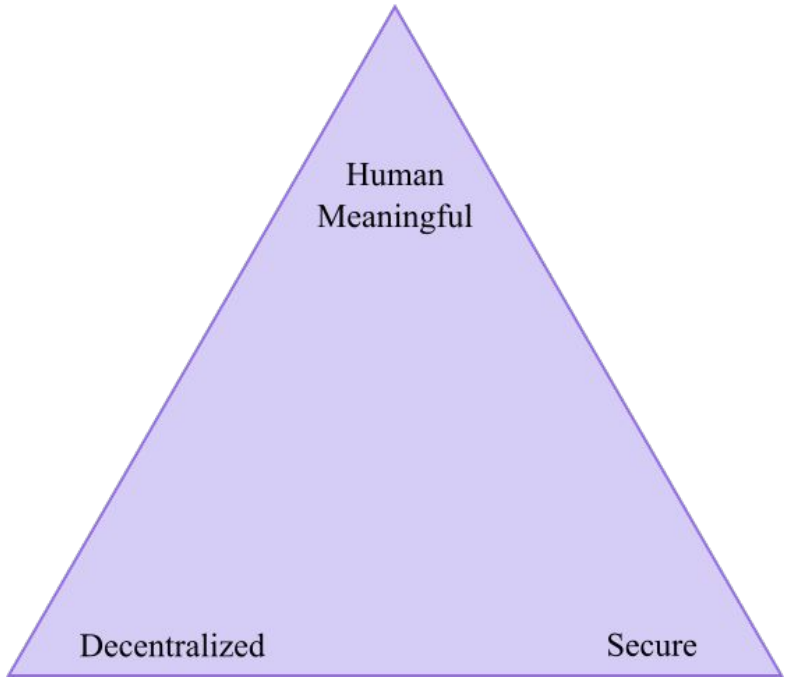
Solving Zooko's Triangle problem

- Zooko's Triangle is a concept in cryptography and naming systems that highlights a trade-off between three essential qualities that are often sought after in naming systems. These three qualities are:
  - **Human-Meaningful:** Easy for people to read and remember.
  - **Secure:** Resilient to manipulation or hijacking.
  - **Decentralized:** No central authority controls it.

- ENS effectively balances all three qualities of Zooko's Triangle:
  - **Human-Meaningful:** ENS names, like vitalik.eth, are easy to remember and user-friendly, eliminating the need for complex blockchain addresses.
  - **Secure:** Built on the Ethereum blockchain, ENS ensures security and immutability, making names tamper-proof once registered.
  - **Decentralized:** ENS operates on Ethereum's decentralized network, removing central authority, which ensures censorship-resistance and greater reliability.



Source: https://support.ens.domains/en/articles/7900404-what-is-ens

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ENS structure and key features

- ENS names are stored as ERC-721 non-fungible tokens (NFTs), with future plans for ERC-1155 compatibility, allowing for more flexible and scalable token standards.
- ENS functions similarly to the Domain Name System (DNS), but operates on the blockchain, offering the same concept of mapping human-readable names to specific resources (e.g., wallet addresses, content, or smart contracts).
- A well-known example of ENS in action is Vitalik.eth, which is a human-readable name that resolves to a specific Ethereum address (0xd8da6bf26964af9d7eed9e03e53415d37aa96045).
- **Programmable Records:** ENS names are flexible and can store various types of records, such as cryptocurrency wallet addresses (e.g., Bitcoin, Ethereum), content hashes, social media handles, and more.
- **Cross-Chain Naming:** ENS is not limited to Ethereum; it supports naming for multiple blockchains and can be used to resolve names for various decentralized applications (dApps).
- **DNS Compatibility:** ENS can import DNS names (such as .com domains) into its system, allowing blockchain-based functionality for existing internet domain names.

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ENS and the blockchain

- ENS operates on Ethereum's decentralized network of smart contracts.
- Unlike DNS, which relies on a series of centralized root servers, ENS uses smart contracts to resolve names to addresses and other records.
- The decentralized nature of ENS means that ENS names and records are not vulnerable to censorship or control by any single entity.
- The **ENS DAO** enables community governance, where changes to the protocol and system are decided through decentralized voting. The ENS DAO Constitution ensures that ENS names cannot be infringed upon or censored, providing users with secure and reliable name ownership.

ENS DAO

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ENS use-cases (1/4)

- **ENS can be used as an easy-to-remember wallet address:**
  - The primary and most common use case for ENS is to simplify cryptocurrency address management.
  - Instead of remembering long, complex hexadecimal wallet addresses, users can use a human-readable ENS name.
  - Supports Multiple Cryptocurrencies: ENS allows users to link multiple cryptocurrency addresses (e.g., Bitcoin, Ethereum) to a single ENS name, enabling multi-crypto payments.
  - This removes the need for users to recall different wallet addresses, making transactions easier

- **ENS names are not limited to wallet addresses. They can store a variety of records such as:**
  - Social media handles
  - Profile pictures
  - Custom records
  - Decentralized website links

- **Web3 Identity:** ENS allows users to create a unified Web3 identity that can integrate across various decentralized applications (dApps) and platforms.

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ENS use-cases (2/4)

- **ENS names can store metadata such as:**
  - An E-mail address
  - A Website URL
  - Social media usernames (e.g., Twitter, Instagram)
- This metadata can be retrieved by dApps and services, eliminating the need to maintain multiple profiles across various platforms.
- This makes it easy for others to find your online presence, linking various digital identities in one place.
- **ENS as a Social Media Identity:**
  - A growing use case for ENS is as a social media identity, linking an ENS name to various social media profiles.
  - Users can attach social media accounts (Twitter, Instagram, etc.) to their ENS names, consolidating their digital identity.
  - How-to: Guides are available for adding social media profiles (e.g., setting a Twitter handle through ENS).

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ENS use-cases (3/4)

- **Several projects allow ENS-based email communication:**
  - ETHMail
  - EtherMail
  - Skiff Mail

- **Custom Service Keys with ENS:**
  - ENS names can also store custom records that can be linked to nearly any kind of service or data.
  - Users can create service keys for various applications or services they interact with.

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ENS use-cases (4/4)

- **Decentralized Website Using ENS:**
  - ENS enables users to point their names to websites hosted on decentralized networks, thus creating fully decentralized websites.
  - This gives users full control over both the domain and hosting, making the website censorship-resistant.
  - Users can set up decentralized websites linked to their ENS names.

IPFS	IPFS (the InterPlanetary File System) is a peer-to-peer network and protocol designed to make the web faster, safer, and more open.
swarm	Swarm is a decentralized data storage and distribution technology. Ready to power the next generation of censorship-resistant, unstoppable, serverless dapps.
a	The Arweave network is like Bitcoin, but for data: A permanent and decentralized web inside an open ledger.
Tor	Tor, short for "The Onion Router," is free and open-source software for enabling anonymous communication.

ENS Supported Decentralized Protocols
Source: https://support.ens.domains/en/articles/7900302-ens-use-cases

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3. Proof of Humanity

The problem of trust online

"Trust" has always been a problem, but has increasingly worsened as our lives moved online:

- The internet allows anyone to create accounts and interact anonymously.
- Fake accounts and bots make it hard to distinguish real people from fraudulent actors.
- This lack of trust erodes the reliability of online systems, from social media to financial platforms.

Some indicative examples:

- Hackers compromised 130 high-profile accounts, including Elon Musk and Barack Obama. (link)
- Facebook removed 1.3 billion fake accounts in the first half of 2023. (link)
- Bots and fake profiles were used to influence voters during the 2016 and 2020 U.S. elections.

Sybil attacks:

- A single individual creates multiple identities and uses them to manipulate systems that rely on unique identities, such as voting, reputation systems, or decentralized networks.

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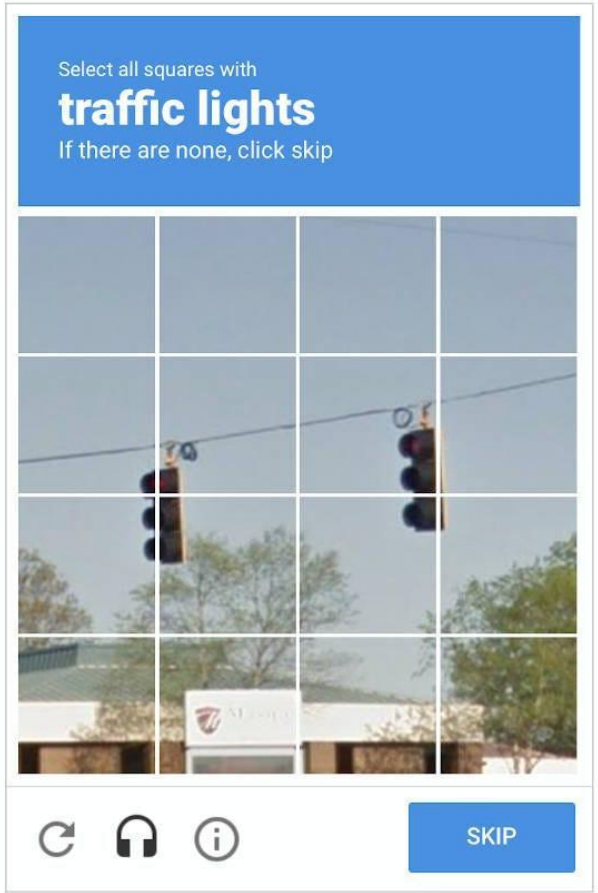
Limitations of traditional verification methods

- Early attempts to mitigate bots, spam, and sybil attacks include:
  - CAPTCHA Tests selecting images or solving puzzles
  - Email or Phone Verification
  - Social Media Verification
  - Knowledge-Based Authentication, e.g., “What is your mother’s maiden name?”
  - IP Address Checks

- However, they all face similar limitations, including:
  - **Vulnerability to Automation:** Advanced bots and AI models can bypass many verification mechanisms (e.g., solving CAPTCHAs or mimicking human behavior).
  - **Privacy Concerns:** Methods like email/phone verification or KBA expose sensitive personal data.
  - **Exclusion Issues:** People without access to certain technologies or with accessibility challenges are unfairly excluded.
  - **Centralization Risks:** Relied on centralized authorities, making systems prone to single points of failure and censorship.



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Consensus mechanisms are a form of sybil resistance

Even consensus mechanisms were created as a form of sybil resistance.

- For instance, Bitcoin introduced PoW a s a “one CPU, one vote” system.
- The idea was that participation would be tied to some minimal hardware requirement, that would allow anyone to participate but make it costly to exploit the system.
- The “one CPU” can be thought of as a proxy for individual participation, or identity.

However, despite PoW being decentralized, **it still cannot serve as a Proof of Humanity or Personhood**, as there is no mechanism to ensure that a single individual is limited to operating only one node.

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Introducing Proof of Humanity or personhood

Proof of Humanity systems are designed to verify that a digital identity corresponds to a real human being. They combine blockchain technology with community-driven validation to ensure:

- **Uniqueness**: Each identity is tied to a single individual.
- **Decentralization:** By removing reliance on a single authority, it reduces vulnerabilities like censorship and single points of failure, creating a more resilient system.
- **Security:** Blockchain's immutable and tamper-proof nature ensures that identity records cannot be altered, enhancing trustworthiness.
- **Scalability:** The community-driven model enables these systems to grow organically, adapting to diverse global contexts and needs without bottlenecks.
- **Privacy:** By using cryptographic methods, individuals can prove their humanity without revealing sensitive personal data, safeguarding their privacy in an increasingly surveilled digital environment.
- **Global Accessibility:** Blockchain's borderless nature and the inclusiveness of community validation allow participation from individuals in underserved regions, empowering those without traditional forms of identification.

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Approaches to achieving Proof of Humanity

- **In-Person Verification Events:** Participants gather in physical locations to verify each other. Pseudonymous participation can ensure unique identities without formal documentation.

- **Social Network-Based Verification**: Individuals vouch for each other within decentralized networks. Web of Trust models build identity through mutual attestations, and systems can enhance verification with cryptographic proofs.

- **Biometric Verification Systems:** Biometric methods link unique identifiers to decentralized records. Privacy-preserving techniques ensure data stays local, while blockchain integration prevents duplication and enhances security.

- **Task-Based Verification Systems:** Validation networks rely on collaborative human tasks to confirm identities. These systems are more complex than reCAPTCHAs and make it difficult for bots or fake accounts to manipulate verification processes.

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The Worldcoin identity system

- Worldcoin is a project aiming to create unique digital identities using iris scans to prevent duplication and establish Proof of Humanity.
- Users visit an Orb device that scans their iris and generates a unique, encrypted World ID, stored on the blockchain.
- The goal is to distribute UBI (Universal Basic Income) to humans verified via the orb, however the economics of the project and their scalability to mass adoption are not well defined.

- There are also numerous other limitations associated with this approach, to name a few:
  - **Privacy Concerns**: Critics argue that even with encryption, biometric data collection poses risks.
  - **Accessibility**: Requires physical access to an Orb, which may exclude users in remote areas.
  - **Centralization Risk**: The reliance on proprietary Orb introduces potential points of failure or control.

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4. Introduction to DAOs

Defining a Decentralized Autonomous Organization (DAO)

In the literature the definition of a DAO varies. A good way to define a DAO was published in 2023 by Rikken, Janssen & Kwee:

A DAO is a system in which storage and transaction of value and notary (voting) functions can be designed, organized, recorded, and archived, and where data and actions are recorded and autonomously executed in a decentralized way.

As of November 2024, the combined market capitalization of all DAOs is approximately $27.4 billion. This valuation reflects the total worth of DAO-related cryptocurrencies and tokens, highlighting the significant role DAOs play in the broader cryptocurrency ecosystem.

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Why do we need a DAO - what is **governance** anyways?

Governance is a framework of rules and procedures that regulates conduct of participants of a network (or corporation)

Feature	Traditional Governance	DAO / Blockchain Governance
Decision-making	Centralized (e.g., board members)	Decentralized (community vote)
Enforcement	Human intermediaries	Smart contracts (“code is law”)
Accountability	Legal and regulatory frameworks	It depends…
Flexibility	Dynamic (human judgment)	Rigid (defined by code)

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Governance: From Tradition to Decentralization (1/2)

Traditional Governance (Centralized Decision-Making)

↓

Web 3.0 & Blockchain (Reduced Intermediaries, Transparency)

↓

DAO Governance ((in theory) Fully Decentralized, Autonomous)

Traditional Governance:

Key Features:

- Centralized authority (boards, executives).
- Hierarchical structure.
- Decisions enforced by intermediaries or regulatory bodies.

Example:

- Corporate governance with shareholder voting and board decisions.

Rising demand for transparency, inclusivity, and reduced reliance on intermediaries.

Blockchain-enabled trustless systems as enablers.

DAO Governance:

Core Principles:

- Decentralized decision-making.
- Self-executing smart contracts.
- Community-driven participation through governance tokens.

Example:

- MakerDAO enabling community governance of collateralized lending protocols.

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Governance: From Tradition to Decentralization (2/2)

Decentralized governance represents an evolution towards inclusivity and autonomy, yet it brings new challenges requiring innovative solutions.

JANINA WINTER MAR 11, 2024

Whales, Sybil attacks and low trust: Can DAOs avoid centralization pitfalls?

DAOs promise collective and transparent decision-making. However, whales and fake accounts threaten fair voting.

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HOME < NEWS < 0XRESEARCH NEWSLETTER < DEFI

The DAO dilemma: Striving for decentralization

While DAOs promote transparency and community involvement, underlying centralized mechanisms reveal the challenges in achieving full decentralization

BY DONOVAN CHOY / NOVEMBER 4, 2024 01:00 PM

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Governance Tokens

- Decentralization of participation and governance is at the heart of blockchain protocols and DeFi applications. To achieve it, many DeFi projects issue **governance tokens** that aim to decentralize the governance of the underlying protocol.
- **Governance tokens** can be:
  - **Earned**, by participating in certain activities, such as providing liquidity (think back to week 4).
  - **Minted**, for protocol creators and early investors.
  - **Gifted**, to addresses that fulfil specific criteria in a process known as an **airdrop**.
  - **Bought and sold**, in DEXes and CEXes, as with any other publicly traded token.
- Governance tokens are not to be confused with **utility tokens**, which can be exchanged for specific services.
- Governance token holders collectively decide on the future of a project by organizing themselves in **Decentralized Autonomous Organizations (DAOs)**.

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Why is the question of governance relevant in DeFi?

There are two reasons why governance in DeFi is important:

1. It puts the “De” (decentralization) in DeFi
2. It is a necessary consequence of how value is managed by applications deployed on higher DeFi stack layers

In the words of Vitalik Buterin:

> “*for application-layer projects, and especially DeFi projects,* ***we run into the problem that application-layer smart contract systems often directly control external assets****, and that control cannot be forked away.*
>
> *If Tezos's on-chain governance gets captured by an attacker,* ***the community can hard-fork away without any losses*** *beyond (admittedly high) coordination costs.*
>
> *If MakerDAO's on-chain governance gets captured by an attacker,* ***the community can absolutely spin up a new MakerDAO****,* ***but they lose all the ETH*** *and other assets that are stuck in the existing MakerDAO CDPs.*
>
> *Hence, while off-chain governance is good for base layers and some application-layer projects,* ***many application-layer projects, particularly DeFi, will inevitably require formalized on-chain governance***”

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An Introduction to DAOs (1/2)

Bitcoin gave birth to the notion of **unstoppable money** and **Ethereum** to the notion of **unstoppable code**.

DAOs are **unstoppable money + unstoppable code put together to create unstoppable organizations.**

To get an understanding of what a DAO is, let's examine the acronym. DAOs are (in theory):

- **Decentralized**: They do not possess a physical address and may not be, in some jurisdictions, directly subject to the established rules and procedures regarding corporations. Instead, they are entirely incorporated online, and live on the blockchain, upheld by the nodes that help maintain the network.
- **Autonomous**: They are completely defined by self-executing code (smart contracts) and operate under the principle of "code is law", within the limitations of the blockchain infrastructure that they are deployed upon.
- **Organization**: Meaning a collection of entities that come together to fulfill a common goal.
  - DAOs are agnostic to who or what interacts with the network and to the reasons they send or receive information or value. All they care about is receiving and sending valid information, as defined by the rules of consensus.
  - This means that blockchains, and by extensions DAOs can interact with humans, AI, or anything that can send and receive valid information.
  - As Richard Gendal Brown put it: On the blockchain nobody knows you are a fridge

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An Introduction to DAOs (2/2)

- Some argue that DAOs **are not directly comparable** to traditional organizations that are built to optimize for profit, often at the expense of the collective benefit, producing severe externalities and operating at the expense of inclusion, sustainability, resilience.
  - For example, they can inherit many of the properties of their underlying blockchain networks and thus be permissionless, decentralized, trustless, transparent, censorship resistant – or not

- Others argue that an ideal DAO would be **value-driven,** governed entirely by its individual members who democratically make decisions about the future of the project.
  - In practice, as with regular organizations, **DAOs are neither perfectly decentralized or centralized, nor perfectly "good" or "evil"**. They are influenced by the choices of their members and the environment they operate in.

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Governance through DAOs

- DAO governance models present many similarities with traditional corporate governance.
- Governance token holders can:
  - **Create governance proposals** about the future of the protocol
  - **Vote** on proposals
- Proposals that reach some level of consensus (as defined in the smart contracts) are accepted and enforced.
- However, as we will discuss in the following slides, DAOs go beyond governance.

***Buying governance tokens from a centralized exchange:** Buying governance tokens from a centralized exchange means you **can't vote directly**, as the tokens are owned by the exchange, which may vote on your behalf. However, if you **transfer the tokens to your personal wallet**, you gain full control and **can participate in governance decisions** by voting directly.

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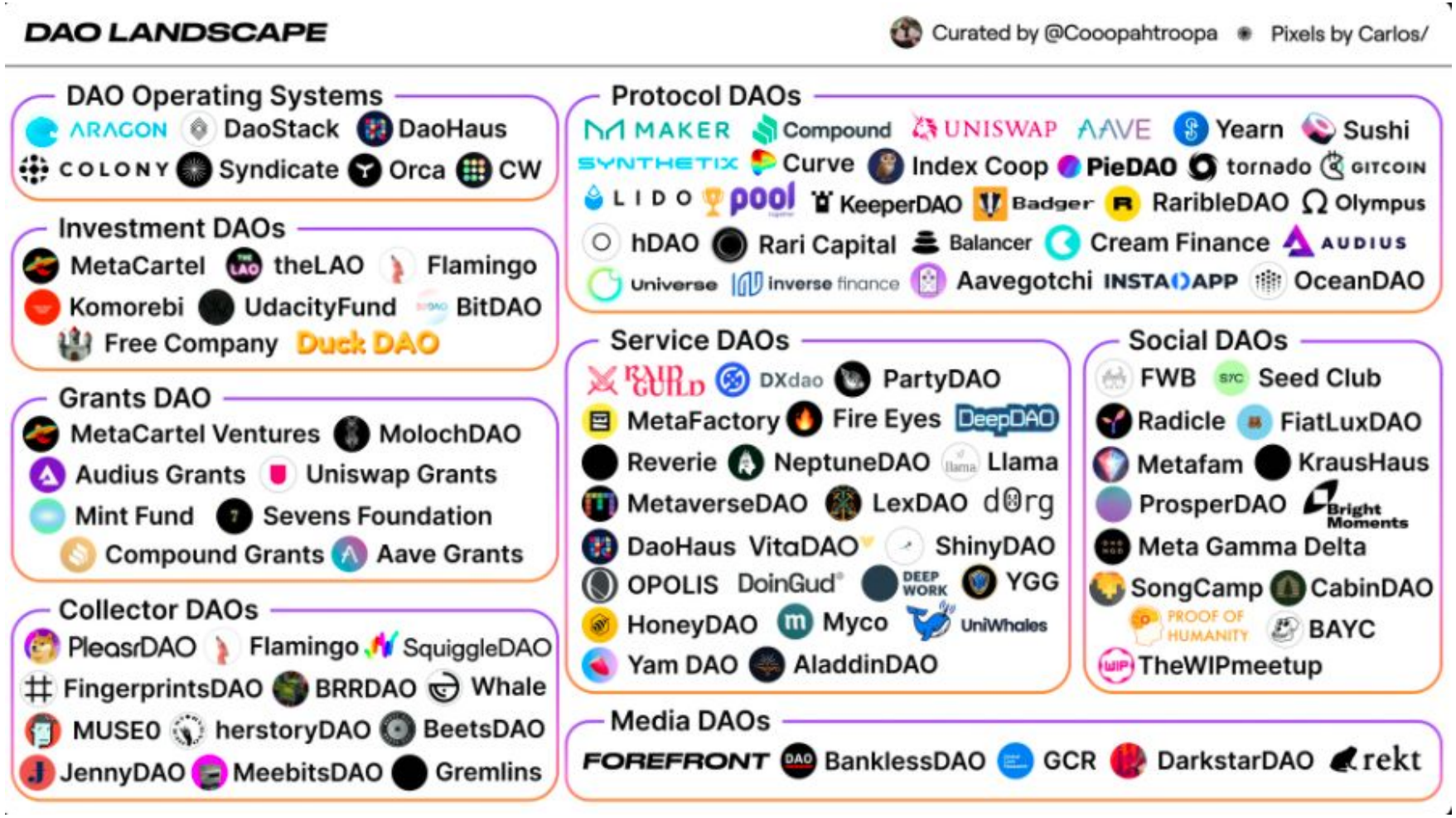
Session: DAOs, Decentralized Social Media, and Digital Identity

5. DAO Examples and Characteristics

DAOs come in many forms



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DAO Examples (1/4)

Uniswap:

- Uniswap is a decentralized exchange (DEX) that allows users to trade cryptocurrencies directly from their wallets using **automated liquidity pools** instead of traditional order books.
- Its **DAO governance model** includes the proposal and vote system for its governance token. For users to submit a proposal for the platform, they must own at least 1% of the total issued governance token UNI.

UNISWAP

Compound:

- Compound DAO governs the Compound protocol, a decentralized finance platform that enables users to lend and borrow cryptocurrencies through algorithmically determined interest rates, with governance decisions made by COMP token holders.
- Users can transfer their governance tokens (COMP) to other users to vote (Governance-as-a-Service). Compound also enforces a minimum 1% stake to submit a proposal.

Compound

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DAO Examples (2/4)

Aave:

- Aave DAO governs the Aave Protocol, a decentralized non-custodial liquidity platform where users can supply and borrow assets, with AAVE token holders proposing and voting on protocol changes.
- The governance token AAVE (and stkAAVE) provides its holders with proposal and voting power.

MakerDAO:

- MakerDAO is a decentralized platform that enables users to create and manage the DAI stablecoin, which is pegged to the US dollar and backed by various types of cryptocurrency collateral.
- **Note:** MKR holders vote on adding collateral types, adjusting stability fees, and governance improvements. For example, the inclusion of real-world assets as collateral was a community-driven decision.
- Participants interested in voting must lock-up their MKR (Maker DAO's governance token) by transferring them from their private wallets to the voting contract, from where they can be withdrawn at any time.

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DAO Examples (3/4)

Curve:

- Curve DAO **governs Curve Finance**, a decentralized exchange optimized for low-slippage swaps between stablecoins and similar assets, utilizing automated market maker (AMM) algorithms to provide efficient trading and liquidity provision.
- The Curve DAO also uses **proposal/voting methods to upgrade the protocol**. For different lengths of time (minimum one week; maximum four years) CRV holders can lock their tokens so they can obtain vote escrowed CRV (veCRV) to gain voting and proposal power. There is no voting power minimum, but a user must have 2,500 veCRV to create a proposal.

Curve

Synthetix:

- Synthetix DAO governs Synthetix, a decentralized protocol for issuing and trading synthetic assets (synths) that track the value of real-world assets like currencies, commodities, and stocks, enabling permissionless derivatives trading while managing treasury and ecosystem incentives through decentralized governance.
- The **SNX token** is used as collateral for minting synthetics. Proposed improvements can be issued by anyone. Yet, approval only comes from the Spartan Council which consists of seven annually elected individuals that review proposals, interview proposal authors and implement approved proposals. Additional councils, such as the Treasury Council and Grants Council, oversee specific areas like financial management and ecosystem development.

SYNTHETIX

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DAO Examples (4/4)

Yearn Finance:

- The Yearn Finance DAO **governs a suite of decentralized finance products** that optimize yield generation on digital assets through automated strategies and community-driven decision-making.
- In April 2021, Yearn Finance implemented Governance 2.0, introducing **a multi-DAO framework** managed through constrained delegation. This structure comprises various independent groups, including YFI token holders, specialized teams known as yTeams, and a Multisig committee. YFI holders possess ultimate authority, voting on protocol changes and governance structures. yTeams focus on specific protocol aspects, while the Multisig executes or vetoes on-chain decisions. This arrangement aims to balance decentralization with operational efficiency.

Rarible:

- Rarible is a decentralized NFT marketplace that enables users to create, buy, and sell unique digital assets across multiple blockchains, fostering a community-driven platform for digital collectibles and art.
- Rarible employs a vote-based governance model where users earn RARI tokens through active participation on the platform. Holders of RARI tokens can submit proposals and vote on platform upgrades, curation, and moderation decisions, thereby influencing the platform's development and direction.

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DAOs beyond governance

There are many uses of DAOs, beyond governance, of course…

For instance, many are starting to realize the benefits that come with working for flexible, internet-native organizations. DAOs utilize a plethora of tools and mechanisms to compensate and reward, not only core, but also occasional contributors. These may include:

- Full employment, like in a traditional business.
- Grants for specific projects.
- Bounties for uncovering bugs/vulnerabilities and exploits.
- Other revenue sharing through peer assessment programs.

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The Key Properties of a DAO: Autonomy and Decentralization

- **Autonomy:** The concept of autonomy varies by context and has different meanings to different disciplines:
  - **Social Sciences:** Autonomy **traditionally** refers to independence from a central authority, emphasizing self-governance in decision-making.
  - **Philosophy (e.g., Kant):** Focuses on autonomy of the mind and consciousness, emphasizing moral and intellectual independence.
  - **Computer Science & Engineering:** Autonomy is defined as a system's capacity to adapt and function effectively under uncertainty.

> With recent advancements in AI and robotics, these scientific disciplines now dominate both research discussions and media coverage.
> *Note: An "if-then-else" statement is not true autonomy; it merely reacts to predefined conditions rather than uncertainty.*

- **Decentralization:** Decentralization involves distributing decision-making authority across a broad network of participants rather than concentrating it in a central authority.
  - This is achieved through mechanisms like **token-based voting**, **smart contracts**, and **community governance**, which empower stakeholders to shape the organization's rules, operations, and evolution.
  - Decentralization ensures inclusivity, transparency, and resilience within the DAO structure.

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So why does it all matter? (1/2)

- The **Tornado Cash DAO governs Tornado Cash**, a decentralized privacy protocol on Ethereum that enables anonymous cryptocurrency transactions using zero-knowledge proofs.
- Governance decisions, such as protocol upgrades and parameter changes, are made by holders of the **TORN token**. Token holders can propose and vote on changes, with voting power proportional to the amount of TORN held.
- In August 2023, U.S. authorities charged Tornado Cash founders Roman Storm and Roman Semenov with **conspiracy to commit money laundering**, sanctions violations, and operating an unlicensed money-transmitting business.
- The indictment alleged that they facilitated the **laundering of over $1 billion in criminal proceeds** through their cryptocurrency mixing service, including funds linked to the North Korean cybercrime group Lazarus. These actions were deemed to violate U.S. sanctions and anti-money laundering regulations.

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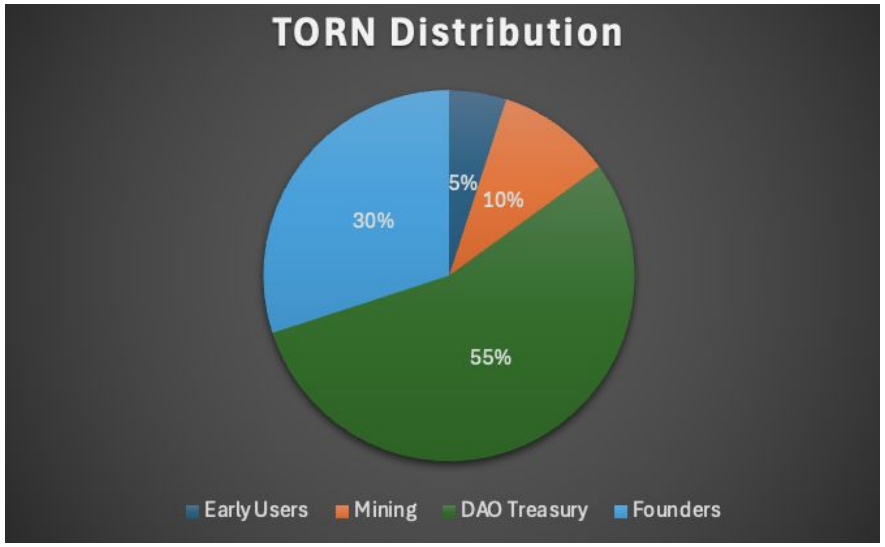
So why does it all matter? (2/2)

- Tornado Cash DAO was neither fully autonomous nor truly decentralized. A significant portion of TORN tokens (55%) was allocated to the DAO treasury, while 30% went to the founding developers and early supporters, concentrating governance power within a limited group.
- If a DAO is truly autonomous in an ethical sense, it can be considered a moral agent and should be held accountable for unethical behavior. However, if it is not autonomous in the true sense of morality, then who is responsible for the actions of a DAO?" (Tan et al., 2024, Open Problems in DAOs)
- From a legal perspective, such a DAO is classified in most jurisdictions as a partnership, exposing token holders to full personal liability.



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Future directions and open challenges

- **Interdisciplinary Nature:** DAOs exist at the intersection of social science and computer science, blending technology with human-centered experimentation.

- **Research Perspectives:**
  - **Computer Science:** Privacy concerns, secure voting, human-computer interaction, multi-chain DAOs, and system design.
  - **Economics:** Measuring governance costs, labor dynamics, and principal-agent problems.
  - **Ethics:** Exploring DAOs as moral agents and their responsibility for unethical behavior.
  - **Law:** Legal uncertainties, liability, bankruptcy, and regulatory challenges.
  - **Political Science:** Power dynamics, voting mechanisms, and digital governance models.

- **Emerging Applications:**
  - Governing AI
  - **Regenerative Finance:** Focusing on sustainability and fair profit distribution.

DAOs are experimental platforms that provide insights into governance, autonomy, and coordination, while also posing significant challenges in privacy, ethics, law, and economics.

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Session: DAOs, Decentralized Social Media, and Digital Identity

6. Decentralized Social Media

Challenges of Centralized Social Media

Data Ownership and Privacy:

- Centralized platforms often control user data, leading to privacy concerns and potential misuse.
- Frequent data breaches highlight vulnerabilities in centralized data storage.

Content Moderation:

- Opaque and inconsistent moderation policies can result in censorship or the spread of misinformation.

Monetization and Exploitation:

- Creators may receive minimal compensation, while platforms retain significant profits.

Centralized Power:

- A few entities act as gatekeepers, influencing public discourse and user behavior.

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Advantages of Decentralized Social Media

User Control and Ownership:
- Users have greater control over their data, enhancing privacy and autonomy.

Transparent Governance:
- Decisions are made collectively, reducing bias and increasing transparency.

Resilience Against Censorship:
- Decentralized architectures make it harder for any single entity to censor or manipulate content.

Fair Monetization:
- Blockchain-based incentives allow creators to earn directly from their audience without intermediaries.

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Centralized vs. Decentralized Social Media

Feature	Centralized	Decentralized
Data Ownership	Platform	User
Governance	Platform	Community (User)
Content Moderation	Opaque and inconsistent	Transparent but complex (scalability issues)
Censorship	Single-point control	Resistant / democratic
Monetization	Platform	Blockchain incentives
Scalability	High	Limited due to technology and adoption

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Challenges of Decentralized Social Media

Scalability:
- Decentralized networks often struggle to handle the volume and speed of traditional platforms.

User Experience:
- Interfaces and functionalities may be less polished compared to centralized counterparts.

Content Moderation:
- Effective moderation is complex, potentially leading to the spread of harmful content or misinformation.

Adoption and Network Effects:
- Centralized platforms dominate due to large user bases, creating barriers for decentralized alternatives.

Regulatory Uncertainty:
- Decentralized platforms often face unclear or restrictive legal frameworks.

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Examples of decentralized social media

- **Mastodon:** A federated network where users control their servers, promoting community-driven moderation and privacy.
- **Bluesky:** Built upon the AT Protocol, Bluesky aims to create a decentralized foundation for public social media, enabling multiple interoperable providers and giving users agency over the content they see.
- **Secure Scuttlebutt (SSB):** A peer-to-peer communication protocol that allows users to host their own content, providing fault tolerance and resistance to censorship.
- **Diaspora:** A decentralized and open-source platform where data is stored on user-owned servers (“pods”), giving users control over their information.
- **Steemit:** A blockchain-based platform that rewards users with cryptocurrency for content creation and curation, promoting fair monetization.

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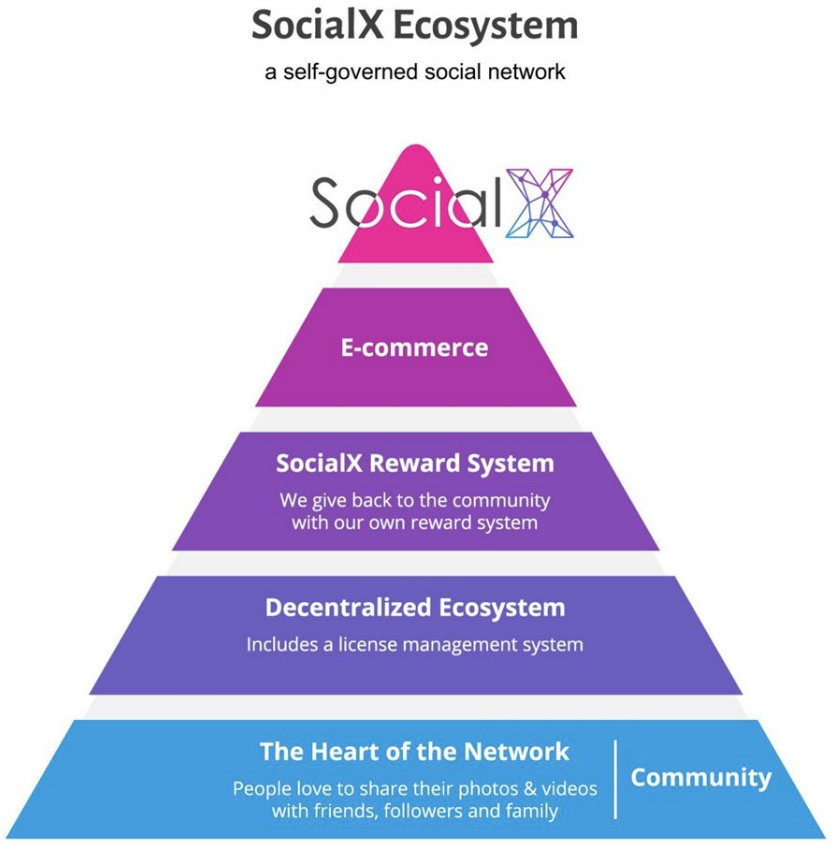
Example of a technology stack - SocialX



Source: https://socialx.network/ecosystem/

- SocialX is a blockchain-based social media platform that combines community building, cryptocurrency rewards, and secure interactions.
- Its mission is to give users ownership of their content, uphold freedom of speech, and reward contributors for their creative work.
- By leveraging blockchain technology, SocialX aims to decentralize social networks, transforming the way people share, interact, and transact online.

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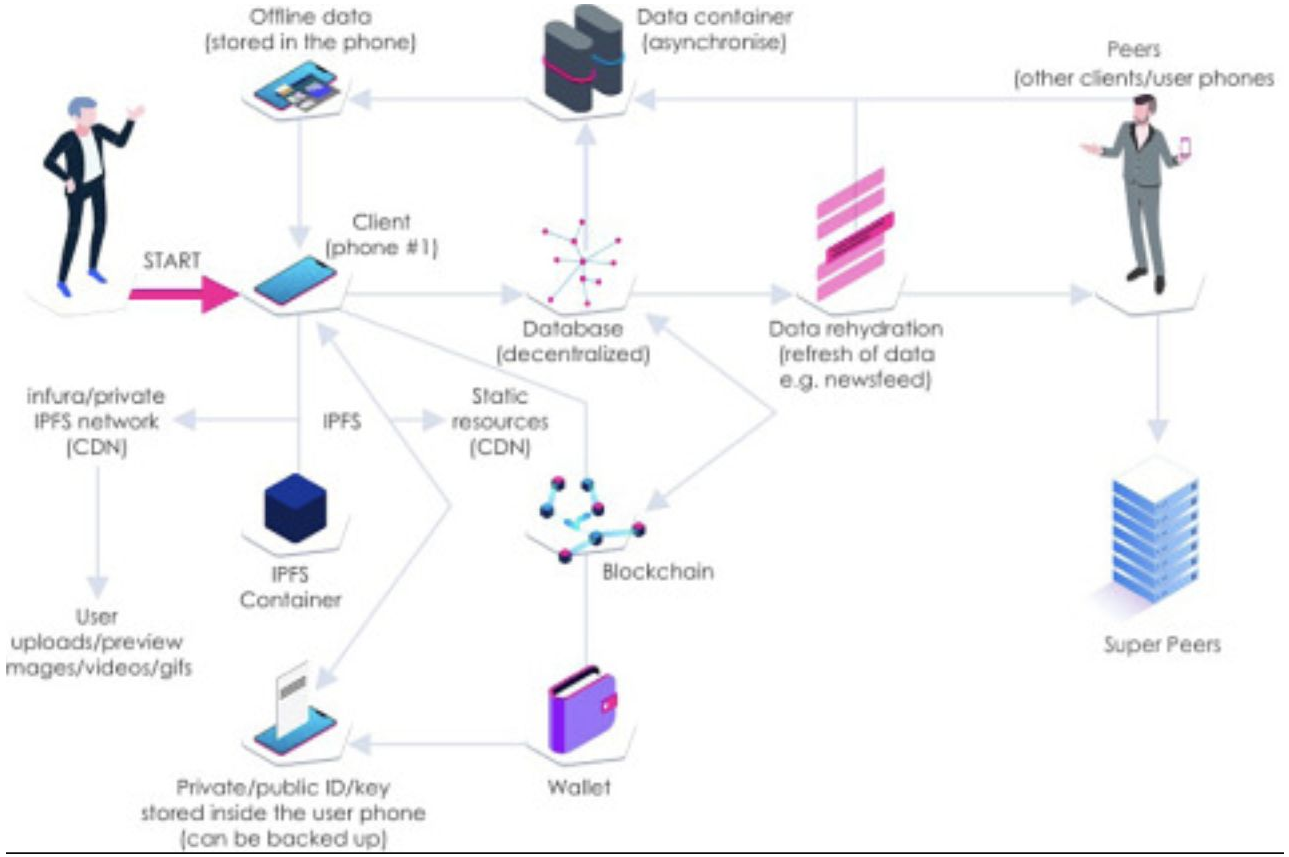
Example of a technology stack - SocialX

1. User uploads content
2. Content gets optimized
3. IPFS is utilized for storage of the content (Infura)
4. Blockchain technology is used for the handling of transactional data and the SocialX app
5. It also provides a p2p network for computation and storage
6. Zero-Knowledge-Proof is implemented for encryption for party verification



Guidi 2020, When Blockchain meets Online Social Networks

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Session: DAOs, Decentralized Social Media, and Digital Identity

7. Conclusions

Conclusions (1/2)

- SSI provides individuals with full control over their digital identity, enhancing privacy and security.
- Blockchain, DIDs, and VCs form the foundation of SSI, ensuring decentralization, security, and data integrity.
- The Trust Triangle (Holder, Issuer, Verifier) ensures secure, privacy-preserving identity management.
- ENS simplifies blockchain address management by translating complex addresses into human-readable names.
- ENS enables decentralized, censorship-resistant naming and supports cross-chain functionality.
- Proof of Humanity systems combine blockchain and community validation to verify real human identities, reducing fraud and enhancing trust.
- DAOs combine social and computer sciences with blockchain to create innovative governance structures.
- While DAOs promise autonomy, transparency, and decentralization, they face challenges such as governance token centralization, privacy issues, and legal uncertainties.
- Research is exploring solutions in secure voting, standardization, and new economic models for DAOs.

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Conclusions (2/2)

- DAOs also act as platforms for testing theories in organizational behavior, political science, and ethics.
- Decentralized social networks give users control over their data, enhancing privacy and resisting censorship. These networks use blockchain for community-driven governance and reward systems for creators.
- Projects like Mastodon, Lens Protocol, and SocialX show the potential to decentralize content ownership and promote freedom of expression.
- Challenges like scalability, user adoption, and moderation remain barriers for decentralized social networks.
- DAOs and decentralized social networks contribute to governance, digital interaction, and economic inclusivity.
- To reach their full potential, these systems must balance technical innovation with ethical, legal, and operational considerations.
- DAOs and decentralized networks are pivotal in driving the next phase of the digital economy.

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8. Further Reading

Further Reading (1/4)

In search of self-sovereign identity leveraging blockchain technology
https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8776589

Self-sovereign identity solutions: The necessity of blockchain technology
https://arxiv.org/abs/1904.12816

The ENS DAO Constitution
https://docs.ens.domains/dao/constitution

Vitalik Buterin on the need for Proof of Humanity
https://github.com/ethereum/wiki/wiki/Problems/89fd07ffff8b042134e4ca67a0ce143d574016bd

Proof-of-Personhood: Redemocratizing Permissionless Cryptocurrencies
https://ieeexplore.ieee.org/document/7966966

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Further Reading (2/4)

An explainer for UBI:
https://www.cnbc.com/2019/06/27/free-cash-handouts-what-is-universal-basic-income-or-ubi.html

World Coin Whitepaper:
https://whitepaper.world.org/

Guaranteed Minimum Income in the Ancient Athenian Constitution
https://classics.mit.edu/Aristotle/athenian_const.3.3.html

European Digital Identity
https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/europe-fit-digital-age/european-digital-identity_en

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Further Reading (3/4)

(Hassan, De Filippi, 2021) Decentralized Autonomous Organization
https://policyreview.info/glossary/DAO

(Li, Qin & Wang, 2023): The Future of Management: DAO to Smart Organizations and Intelligent Operations
https://ieeexplore.ieee.org/abstract/document/9994617/

(Qin et al., 2023) Web3-Based Decentralized Autonomous Organizations and Operations: Architectures, Models, and Mechanisms
https://ieeexplore.ieee.org/abstract/document/9993805

(Feichtinger et al., 2023) The Hidden Shortcomings of (D)AOs -An Empirical Study of On-Chain Governance
https://link.springer.com/chapter/10.1007/978-3-031-48806-1_11

(Austgen et al., 2023) DAO Decentralization: Voting-Bloc Entropy, Bribery, and Dark DAOs
https://arxiv.org/abs/2311.03530

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Further Reading (4/4)

(Ziegler, Dupont, 2023) Navigating the Research Landscape of Decentralized Autonomous Organizations: A Research Note and Agenda
https://arxiv.org/abs/2312.17197

(Ziegler, Welpe, 2022) A Taxonomy of Decentralized Autonomous Organizations
https://mediatum.ub.tum.de/1709396

(Tan et al., 2024) Open Problems in DAOs
https://arxiv.org/abs/2310.19201

(Guidi, 2020) When Blockchain meets Online Social Networks
https://www.sciencedirect.com/science/article/abs/pii/S1574119220300195

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